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CANDIDATE-PLEASE NOTE!

You must sign below and return this booklet with the Answer Sheet. Failure to do so may result in disqualification.

Signature

TEST CODE **01207010**

MAY/JUNE 2008

FORM TP 2008042

CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

BIOLOGY

Paper 01 – General Proficiency

75 minutes

09 JUNE 2008 (p.m)

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. In addition to this test booklet, you should have an answer sheet.
2. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
3. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

Which of the following diseases is due to a dietary deficiency?

- (A) Anaemia
- (B) Diabetes
- (C) Influenza
- (D) Malaria

Sample Answer



The best answer to this item is “Anaemia”, so answer space (A) has been blackened.

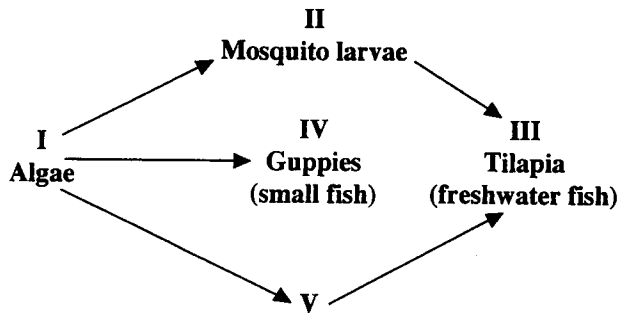
4. If you want to change your answer, be sure to erase your old answer completely and fill in your new choice.
5. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, omit it and go on to the next one. You can return to the omitted item later. Your score will be the total number of correct answers.
6. Figures are not necessarily drawn to scale.
7. This test consists of 60 items. You will have 75 minutes to answer them.
8. Do not be concerned that the answer sheet provides spaces for more answers than there are items in this test.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

1. Which of the following soil organisms can be found living in the root nodules of leguminous plants?

(A) Viruses
(B) Bacteria
(C) Nematodes
(D) Fungi

Items 2-3 refer to the simple web constructed from organisms found in a small pond.



2. Which organism is a predator?

(A) I
(B) II
(C) III
(D) IV

3. The herbivores are

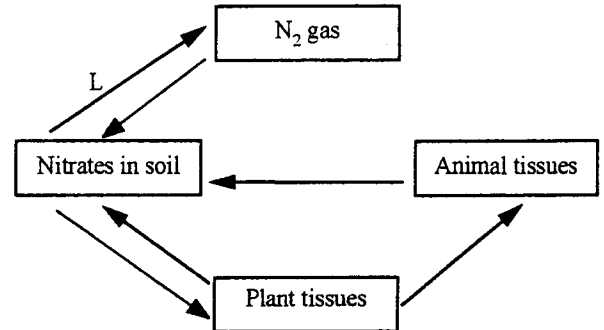
(A) I and III only
(B) II and V only
(C) I, III and IV only
(D) II, IV and V only

4. Some insects are brightly coloured, but are difficult to spot as they sit on plants. This bright coloration may be important for

I. pollination
II. blending into their environment
III. attracting mates
IV. hunting for prey

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II, III and IV

Item 5 refers to the following diagram of the nitrogen cycle.



5. Which of the following groups of bacteria will be active at L?

(A) Dentrifying
(B) Nitrifying
(C) Nitrogen fixing
(D) Putrefying

6. If a carbon atom were traced through the following organisms, the order in which it is likely to be detected would be

(A) small bird → grasshopper → lizard → plant
(B) lizard → grasshopper → plant → small bird
(C) grasshopper → plant → small bird → bird
(D) plant → grasshopper → lizard → small bird

7. In multicellular organisms, cells become different from one another in order to carry out particular functions. This is called

(A) adaptation
(B) selection
(C) specialization
(D) evolution

8. If all the trees in a large area of forest were destroyed by disease, what would be the MOST likely effect on the composition of the air in the long term?

(A) The level of carbon dioxide would decrease.
 (B) The level of carbon dioxide would increase.
 (C) There would be no change in the composition of the air.
 (D) There would be more oxygen.

9. A mature plant cell is different from a mature animal cell because it has

(A) a large permanent vacuole
 (B) mitochondria
 (C) a nucleus suspended in the cytoplasm
 (D) a cell membrane

10. The process which moves food through the alimentary canal is called

(A) digestion
 (B) mastication
 (C) peristalsis
 (D) swallowing

11. Living organisms feed by using simple inorganic substances or by consuming complex organic substances.

Which of the following rows correctly shows the type of substances used or consumed by plants, animals and fungi?

	Plants	Animals	Fungi
(A)	Simple inorganic	Complex organic	Complex organic
(B)	Simple organic	Complex organic	Complex inorganic
(C)	Complex organic	Simple inorganic	Complex organic
(D)	Complex organic	Simple inorganic	Simple inorganic

12. Which of the following would MOST likely occur if enzymes were NOT present in living cells?

(A) Slow rate of reaction
 (B) High rate of reaction
 (C) Reaction requiring higher temperatures
 (D) No reaction

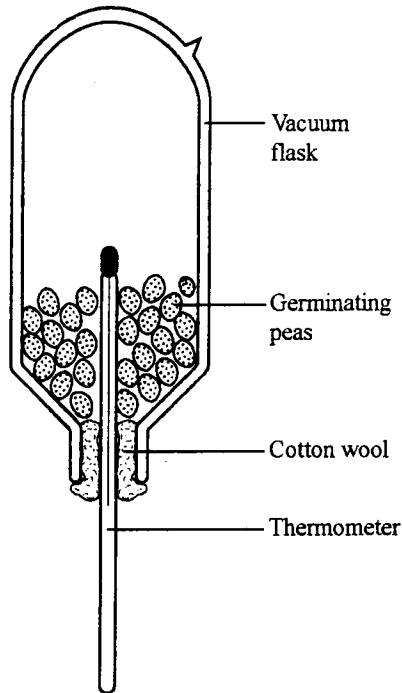
13. Which of the following combinations correctly shows a gas that is a raw material and a gas that is a product, during photosynthesis.

	Raw material	Product
(A)	Oxygen	Nitrogen
(B)	Carbon dioxide	Nitrogen
(C)	Oxygen	Carbon dioxide
(D)	Carbon dioxide	Oxygen

14. Bile empties into the

(A) stomach
 (B) caecum
 (C) ileum
 (D) duodenum

Items 15 - 16 refer to the figure showing an apparatus used to investigate the process of respiration.



15. Which of the following products of respiration is the apparatus used to show?

(A) Heat
(B) Carbon dioxide
(C) Water
(D) Ethanol

16. On setting up the apparatus, bacteria on peas are killed. This is important so that they will not

(A) feed on the peas
(B) give off heat
(C) use up oxygen
(D) use up carbon dioxide

17. The process responsible for the movement of water upwards in xylem vessels is

(A) evaporation
(B) transpiration pull
(C) osmosis
(D) active transport

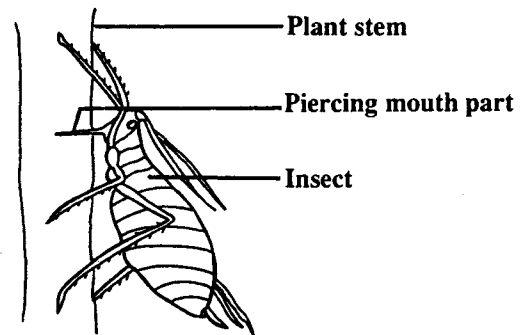
18. Which of the following statements about aerobic and anaerobic respiration is correct?

	Aerobic	Anaerobic
(A)	Does not use O ₂	Uses O ₂
(B)	Occurs in the cytoplasm	Occurs in the mitochondria
(C)	Releases large amounts of energy	Releases small amounts of energy
(D)	At least one product is organic	Products are always inorganic

19. Which of the following comparisons between arteries and veins is NOT correct?

	Arteries	Veins
(A)	Valves present	Valves absent
(B)	Blood flows in pulses	Blood flows smoothly
(C)	Have thick muscular walls	Have thin walls
(D)	Blood low in carbon dioxide	Blood high in carbon dioxide

Item 20 refers to the following diagram of an insect feeding on a plant.



20. The insect MOST likely obtains its nutrients from the

(A) cambium
(B) epidermis
(C) phloem
(D) xylem

21. The molecule used to temporarily store the energy released during the process of respiration is

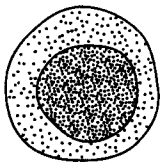
- (A) ADP
- (B) DNA
- (C) ADH
- (D) ATP

22. Which of the following statements about cigarette smoking are correct?

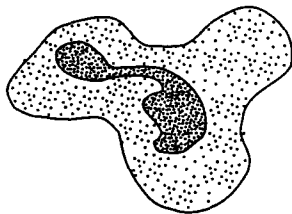
- I. Results in persistent coughing and shortness of breath.
- II. There is destruction of the alveolar walls.
- III. May result in cancer of the lungs, throat, mouth and oesophagus.
- IV. Blood pressure is lowered.

- (A) I and II only
- (B) III and IV only
- (C) I, II and III only
- (D) I, II and IV only

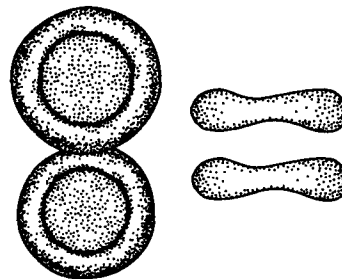
Items 23 - 24 refer to the diagrams below which represent some components of blood.



(A)



(B)



(C)



(D)

In answering items 23 - 24, match each item with one of the options above. Each option may be used once, more than once or not at all.

23. Engulfing disease-causing organisms

24. Transport of oxygen

25. The ureter carries urine from the
- (A) kidney to the bladder
 - (B) collecting tubule to the pelvis
 - (C) nephron to the collecting tubule
 - (D) bladder out of the body

26. While the filtrate passes through the kidney tubule, most of the water moves back into the blood stream. This makes the filtrate
- (A) more dilute
 - (B) darker
 - (C) more concentrated
 - (D) paler in colour

27. Which of the following would NOT represent a form of excretion in plants?

- (A) Water vapour diffusing through epidermis
- (B) Tannins being stored in dead tissue
- (C) Gases diffusing through stomata
- (D) Stored calcium oxalate crystals in leaves during leaf fall

28. Which of the following adaptations in plants is NOT used to conserve water?

- (A) Cuticle thickness
- (B) Long leaves
- (C) Small leaves
- (D) Thorns

29. Which of the following functions are performed by a rib?

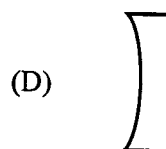
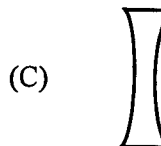
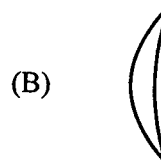
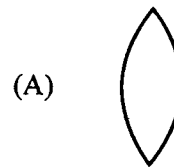
- I. Blood formation
- II. Protection
- III. Locomotion
- IV. Support

- (A) I and II only
- (B) I and IV only
- (C) II and III only
- (D) I, II and IV only

30. The human skeleton can be divided into two parts, the axial skeleton and the appendicular skeleton. Which row in the table groups bones correctly?

	Axial skeleton	Appendicular skeleton
(A)	Pelvic girdle, ribs, femur	Skull, humerus, vertebral column
(B)	Humerus, femur, ribs	Skull, vertebral column, pelvic girdle
(C)	Pelvic girdle, humerus, femur	Vertebral column, ribs, skull
(D)	Skull, vertebral column, ribs	Humerus, femur, pelvic girdle

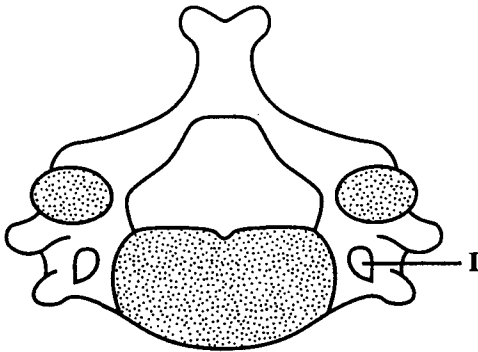
31. Which of the following types of lens will correct short sightedness?



32. Which of the following is NOT involved in the reflex action of the body known as the knee-jerk reflex?

- (A) Spinal cord
- (B) Motor neurone
- (C) Sensory neurone
- (D) Brain

Items 33 - 34 refer to the diagram of the vertebra below.



33. What type of vertebra is represented?

- (A) Cervical
- (B) Lumbar
- (C) Thoracic
- (D) Sacral

34. The structure labelled I allows for the passage of

- (A) blood vessels
- (B) peripheral nerves
- (C) spinal cord
- (D) synovial fluid

35. The following reactions all take place in the skin.

- I. Blood vessels dilate and constrict.
- II. Glands produce sweat.
- III. Hair erector muscles contract.

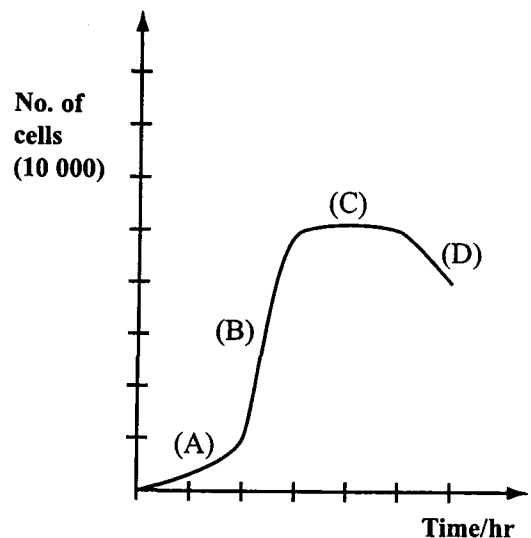
These functions relate to the skin's role as an organ for

- (A) mechanical protection only
- (B) body temperature control only
- (C) both body temperature control and sensitivity to the surroundings
- (D) mechanical protection, body temperature control and sensitivity to the surroundings

36. The eye is sometimes referred to as an effector organ because

- (A) the pupil constricts in the presence of bright light
- (B) the rods and cones are stimulated by light
- (C) images which fall on the retina are sent to the brain
- (D) neurones from the eye lead to the central nervous system

37. The diagram below represents the growth curve of a culture of yeast cells. Which section of the graph shows the MOST rapid rate of growth?



Items 38 - 39 refer to the following information.

The table below gives the glucose content of 5 cm³ samples of blood taken from ten adult patients who were having this test for the first time. Patients were requested not to eat for 12 hours prior to the collection of the sample. The results of the test confirmed that two patients had high glucose levels in the blood, while the blood glucose level was considered within normal limits for the other patients.

Patient	Blood glucose content in mmol/litre	Patient	Blood glucose content in mmol/litre
1	4.0	6	18.0
2	6.5	7	6.0
3	11.5	8	6.5
4	3.5	9	4.0
5	5.0	10	5.2

38. The normal range of glucose in the blood is MOST likely

(A) 3.5 to 18.0
(B) 3.5 to 11.5
(C) 3.5 to 6.5
(D) 3.5 to 4.0

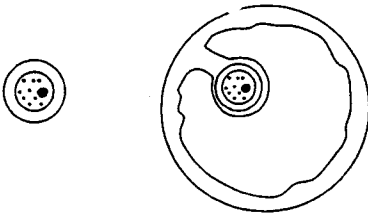
39. The two patients MOST likely suffering from diabetes (high blood glucose) are

(A) 3 and 8
(B) 3 and 6
(C) 2 and 6
(D) 2 and 3

-
40. The reproduction of amoeba by binary fission is an example of asexual reproduction because

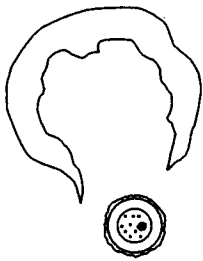
(A) the cytoplasm is divided into two equal halves
(B) two offspring are produced by one parent
(C) the nucleus divides by mitosis
(D) the nucleus is not involved in the cell division

Items **41 - 42** refer to the following diagrams showing events that occur in the ovary during the menstrual cycle.



(A)

(B)



(C)



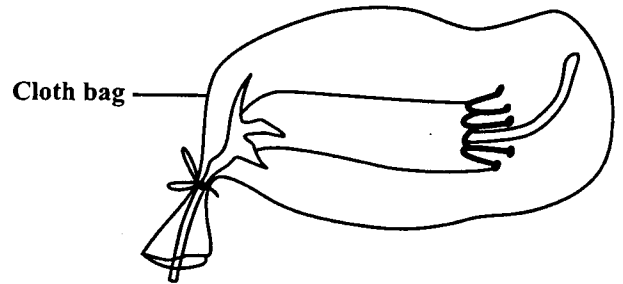
(D)

In answering items **41 - 42**, match each item with one of the options above. Each option may be used once, more than once or not at all.

41. Ovulation

42. Development of corpus luteum

Items **43 - 44** refer to the following diagram which shows part of a flower enclosed in a cloth bag, as part of an investigation being carried out on reproduction in plants.



43. The cloth bag would NOT prevent

- (A) wind pollination
- (B) cross fertilization
- (C) self fertilization
- (D) insect pollination

44. To be effective in controlling how the gametes meet, the bag should be put in place

- (A) after pollination but before fertilization
- (B) before pollination and fertilization
- (C) after fertilization and pollination
- (D) before pollination but after fertilization

45. In a particular plant, red and white flower-colour show co-dominance. If two heterozygous (pink) flowers are crossed, the expected phenotypic ratio of the offspring is

- (A) all pink
- (B) 1 red: 1 white
- (C) 3 red: 1 white
- (D) 1 red: 2 pink: 1 white

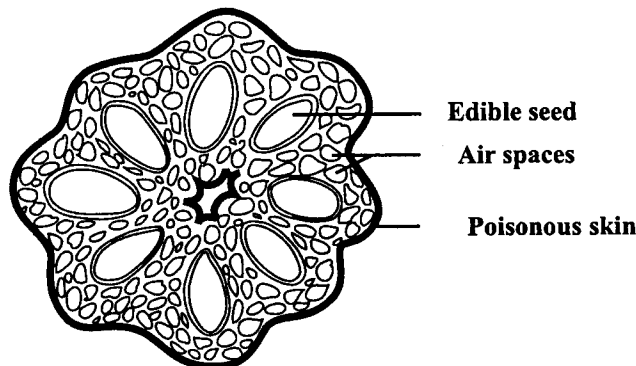
46. Which of the following is an example of continuous variation in humans?

(A) Sex/gender
(B) Tongue rolling
(C) Blood groups
(D) Adult heights

47. Natural selection differs from artificial selection because

(A) the environment determines the individual with the beneficial trait
(B) the environment determines the beneficial characteristics
(C) it is a slower process
(D) it involves non-random mating

Item 48 refers to the following diagram of the cross-section of a fruit of an unknown plant found on the shore of a swamp.



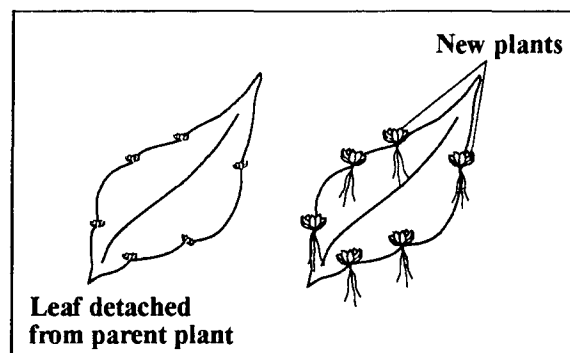
48. Which of the following is MOST likely the method of dispersal?

(A) Animal
(B) Water
(C) Wind
(D) Mechanical

49. If a diploid organism has two different alleles for the same gene, it is described as being

(A) homozygous
(B) recessive
(C) dominant
(D) heterozygous

50. A bryophyllum leaf, isolated from the parent plant, will generate new plants as seen below.



Which of the following is NOT a feature of this type of reproduction?

(A) Plants are generated quickly.
(B) The new plants show great variety.
(C) It is a cheap way to obtain plants.
(D) One plant can generate hundreds of new plants.

51. Which of the following statements are true about natural selection?

I. There is variation in the population.
II. There are more offspring than parents.
III. Better adapted organisms survive and reproduce.
IV. Species would not change over time.

(A) I and II only
(B) II and III only
(C) I, II and III only
(D) I, II, III and IV

52. It is possible to change the characteristics of an organism by inserting pieces of DNA (genes) from one organism into the chromosomes of another. This process is called

(A) cloning
(B) genetic engineering
(C) genetic variation
(D) artificial selection

53. Which of the following would NOT contribute to hypertension (high blood pressure)?

- (A) High salt intake
- (B) Old age
- (C) Regular exercise
- (D) Saturated fats in the diet

54. Immunization is MOST effective in controlling the spread of a communicable disease if

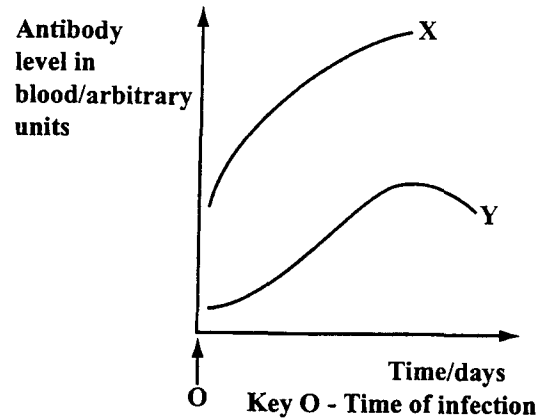
- (A) a vaccine is administered to an uninfected individual
- (B) serum is administered to an uninfected individual
- (C) a vaccine is administered to an infected individual
- (D) serum is administered to an infected individual

55. Which of the following are direct consequences of disease in man?

- I. Loss of man hours and productivity
- II. Loss of human life
- III. Loss of livestock and crops

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

56. The graph below represents the antibody level of two individuals, X and Y, who are infected at the same time with the same pathogen.



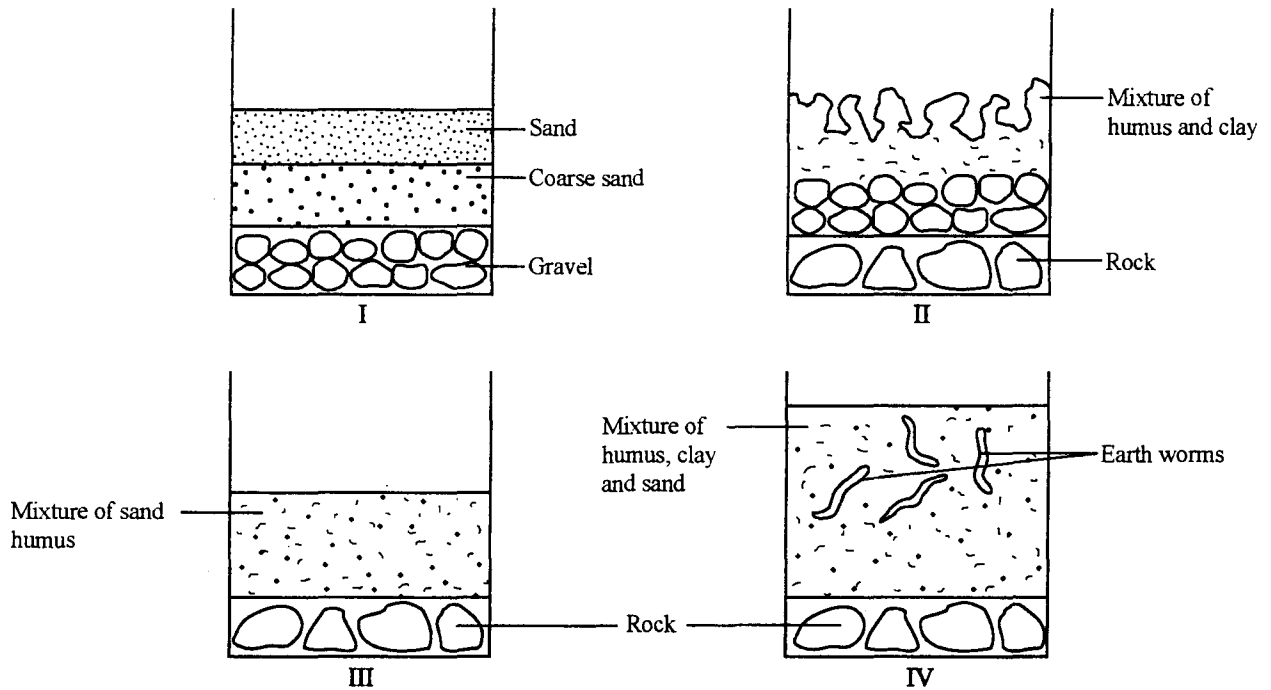
The graph indicates that

- (A) neither X nor Y will ever be sick again
- (B) X has been vaccinated against this disease but not Y
- (C) Y has fewer white blood cells than X
- (D) Y previously had the disease but not X

57. The term 'habitat' refers to

- (A) an area where several organisms live
- (B) the area in which a particular organism lives
- (C) the biological characteristics of an area
- (D) the physical characteristic of an area

Item 58 refers to the diagram below showing four soil samples.



58. In which soil sample will a crop of lettuce grow BEST?

- (A) I
- (B) II
- (C) III
- (D) IV

59. Which of the following are reasons to return glass bottles?

- I. Glass is a non-renewable resource.
- II. Recycling glass lowers production cost.
- III. Raw materials are renewable resources.
- IV. The energy used in bottle making is non-polluting.

- (A) I and II only
- (B) I and IV only
- (C) II and III only
- (D) III and IV only

60. Which of the following is NOT an air pollutant?

- (A) Carbon dioxide
- (B) Lead
- (C) Sulphur dioxide
- (D) Nitrogen